

Algorithm Adaptation for Traffic Queue Analysis

Introduction

The foundational research in "[Real-Time Queue Length Detection with Roadside LiDAR Data](#)" established a method for identifying traffic queues by processing raw, 3D point-cloud data from LiDAR sensors. Our objective is to adapt the core concept of that algorithm—defining queues by identifying slow-moving vehicles—to a new context: the highly structured and pre-processed Waymo Open Motion Dataset. This document details that adaptation.

Waymo Open Motion Dataset

The [Waymo Open Motion dataset](#) provides a rich environment for traffic analysis. Instead of raw sensor readings, it contains a collection of 20-second scenarios in .tfrecord format. Each scenario includes a high-fidelity map detailing every lane, stop line, and road feature. Crucially, it also provides data for all objects on the road, which are already classified (e.g., as vehicles) and tracked over time. The dataset provides each vehicle's exact location, size, and speed across every time step.

Adapted Algorithm

The modified algorithm takes advantage of the structured Waymo data, enabling a simpler and more efficient analysis process. Because the dataset already tells us what an object is and where it's going, we can **bypass the original algorithm's most computationally intensive steps**, such as filtering out background objects, clustering points to identify vehicles, and tracking those vehicles to calculate their speed.

First, for each moment in a scenario, the algorithm scans for all vehicles **moving slower than 5 km/h** (1.4 m/s). There is an initial check to filter out any parked cars. It does this by looking at a vehicle's entire journey within the scenario; if it has **barely moved from its starting spot, it's considered parked** and isn't included in the analysis, which prevents it from being mistaken for a part of a traffic jam.

Next, each of these slow-moving vehicles is assigned to a specific lane. Using the high-definition map, the algorithm calculates **the distance from the vehicle to the centerline of every nearby lane** and assigns it to the closest one. To improve accuracy, this assignment has a tolerance; if a vehicle is more than 5 meters away from the closest lane centerline, it's disregarded. This prevents vehicles in adjacent parking lots or on shoulders from being incorrectly included in a traffic lane.

After assigning all valid slow-moving vehicles to their respective lanes, the algorithm clusters them together. Any lane that contains one or more such vehicles is classified as having a queue. For each identified queue, its main characteristics are calculated. The **queue's length** is measured as the distance from the lane's stop line to the position of the last vehicle in the lineup, and its **density** is determined by the number of vehicles occupying that space.

In the final stage, the algorithm shifts focus to a broader perspective. It uses the map data to understand which lanes are next to each other. If it finds **queues in multiple adjacent lanes** at the same time, it groups them into a larger **"congestion cluster."** This allows us to not only see individual queues but also to identify and report on larger-scale gridlock affecting an entire intersection.

Unique Scenarios

https://drive.google.com/drive/folders/1Kdwu6YPhvB6gFTNTEvRQcuvr2VqlusQ?usp=drive_link

Conclusion

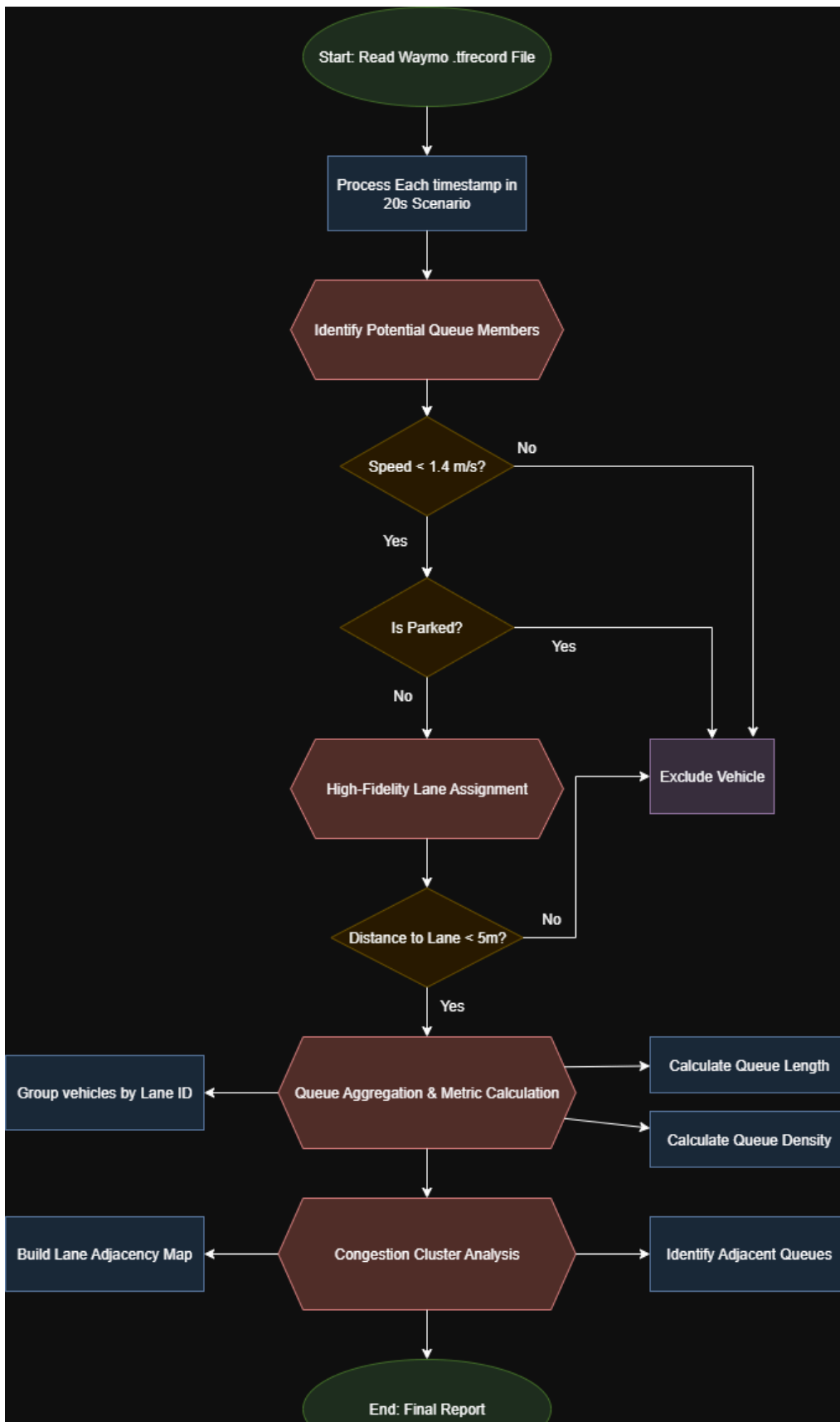
By adapting the original LiDAR-based queue detection algorithm to the structured Waymo Open Motion Dataset, we transform a computationally heavy process into a more direct and scalable solution. The availability of pre-classified and pre-tracked vehicle data allows us to focus on traffic behavior rather than low-level object detection. Our approach reliably distinguishes between parked and slow-moving vehicles, accurately maps them to lanes, and measures both the length and density of resulting queues. Extending this to congestion clusters provides valuable insights into intersection-level traffic flow, enabling a more comprehensive understanding of urban congestion patterns. Ultimately, this adaptation demonstrates how existing queue detection principles can be re-engineered to work with different data environments, paving the way for real-time, data-driven traffic management applications.

Sources

Dataset : Waymo's Open Motion Dataset - [Waymo's Open Motion Dataset](#)

Algorithm :

Wu J, Xu H, Zhang Y, Tian Y, Song X. Real-Time Queue Length Detection with



Working pipeline of the algorithm

